PowerGrid SCADA Network Troubleshooting Guide

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1. SCADA RTU Communication Failure Resolution

1.1 Symptom Identification:
When Regional Load Dispatch Centre (RLDC) operators report "RTU not responding" or "Communication timeout" errors, follow this diagnostic protocol:

Common Error Codes:
• ERR-RTU-001: RTU communication timeout (greater than 5 seconds)
• ERR-RTU-002: Cyclic redundancy check (CRC) failure in data packets
• ERR-RTU-003: IEC 60870-5-104 protocol violation detected
• ERR-NET-101: Subnet mask mismatch in 172.16.x.x network range
• ERR-NET-102: VLAN tagging error (VLAN 100-SCADA, VLAN 200-MIS)

1.2 Layer-by-Layer Diagnostic Procedure:

Step 1: Physical Layer Verification
Required Tools: Fluke DTX-1800 Cable Analyzer, OTDR (Optical Time-Domain Reflectometer)

Procedure:
1. Access the substation RTU cabinet (requires Dual-Lock authentication)
2. Verify fiber optic cable continuity - Acceptable loss: less than 0.3 dB/km for single-mode fiber
3. Check SC/APC or LC/UPC type connectors for damage or contamination
4. Use OTDR to locate fiber breaks (typical detection range: 0-80km)
5. For copper Ethernet (RS-485 legacy systems), verify termination resistor: 120Ω ±5% at both ends
6. Check for ground loops using digital multimeter (maximum distance: 1200m as per IEC standard)

Step 2: Network Layer Troubleshooting
PowerGrid SCADA Network Architecture:
• Primary Network: 172.16.10.0/24 (MPLS-VPN backbone)
• Backup Network: 172.16.20.0/24 (Leased line ISDN)
• Management Network: 10.220.x.x/16 (Out-of-band SSH access)

Commands to execute from SCADA Master Station:
1. ping -c 10 -s 1472 172.16.10.{RTU_IP} # Test MTU size (1500 bytes)
2. traceroute -n 172.16.10.{RTU_IP} # Perform hop-by-hop latency analysis
3. sudo tcpdump -i eth1 port 2404 -vv # Capture IEC 60870-5-104 protocol packets

Expected Network Response Times:
• Intra-region RTUs: less than 20ms latency
• Inter-region RTUs: less than 50ms latency
• Satellite backup link: less than 300ms (acceptable degradation mode)

Step 3: IEC 60870-5-104 Protocol Analysis

Using Wireshark with PowerGrid custom protocol dissector:

Filter: tcp.port == 2404 && iec104

Key Protocol Frames to Monitor:
1. STARTDT (Start Data Transfer) - Should receive STARTDT_CON confirmation within 2 seconds
2. TESTFR (Test Frame) - Heartbeat packet sent every 20 seconds for keepalive
3. ASDU (Application Service Data Unit) - Type ID 1-36 as per Indian Grid Code specification

Common Protocol Issues:
• COT (Cause of Transmission) = 0x03 (Spontaneous) missing indicates RTU not sending status change events
• IOA (Information Object Address) mismatch - Check RTU address mapping in /opt/scada/config/rtu_map.xml
• Timestamp synchronization error - Verify NTP server 10.220.1.100 is reachable from RTU

2. Oracle RAC Database Synchronization Failure

2.1 Problem Statement:
When the hourly cron job (/usr/local/bin/sync_scada_db.sh) fails with exit code 1, and logs show Oracle errors:

• ORA-01017: invalid username/password; logon denied
• ORA-12514: TNS listener does not currently know of service requested in connect descriptor
• ORA-03135: connection lost contact with database server

2.2 Resolution Steps:

Step 1: Oracle RAC Cluster Health Check
Connect to Oracle primary node via secure SSH:
ssh -i /home/scada/.ssh/oracle_key oracleadm@10.220.5.10

Check cluster resource status:
srvctl status database -d SCADA_PROD

Expected output:
Instance SCADA01 is running on node oranode1
Instance SCADA02 is running on node oranode2

If instances are down, execute:
srvctl start database -d SCADA_PROD

Step 2: TNS Configuration Validation
Verify tnsnames.ora file on SCADA master server:
cat /opt/oracle/network/admin/tnsnames.ora

Correct TNS entry format:
SCADA_PROD =
(DESCRIPTION =
(ADDRESS_LIST =
(ADDRESS = (PROTOCOL = TCP)(HOST = oravip1.powergrid.local)(PORT = 1521))
(ADDRESS = (PROTOCOL = TCP)(HOST = oravip2.powergrid.local)(PORT = 1521))
)
(LOAD_BALANCE = yes)
(CONNECT_DATA =
(SERVER = DEDICATED)
(SERVICE_NAME = SCADA_PROD.powergrid.local)
)
)

Test TNS connectivity:
tnsping SCADA_PROD
sqlplus scada_sync/ENCRYPTED_PASSWORD@SCADA_PROD

Step 3: Oracle Wallet Password Management
PowerGrid uses Oracle Wallet for secure credential storage.
Location: /opt/oracle/admin/SCADA_PROD/wallet

Verify wallet status:
mkstore -wrl /opt/oracle/admin/SCADA_PROD/wallet -list

Update credentials (requires DBA approval):
mkstore -wrl /opt/oracle/admin/SCADA_PROD/wallet -modifyCredential SCADA_PROD scada_sync NEW_PASSWORD

Test synchronization script manually:
sudo -u oracle /usr/local/bin/sync_scada_db.sh --dry-run

3. SCADA Password Reset Procedure

3.1 Standard Password Reset Workflow:

Prerequisites:
• Employee badge number in PGCIL-{ZONE}-YYYY-XXXXX format (e.g., PGCIL-NR-2024-12345)
• Zone allocation: NR (North Region), SR (South), ER (East), WR (West), or NER (North-East)
• Registered mobile number for OTP delivery via PowerGrid SecureAuth app
• Confirmation of access to SAP HR portal (transaction code: ZPGHR001)

Step-by-Step Password Reset Process:

Step 1: Identity Verification
IT Support team will verify the following in HRMS database:
• Employee badge number matches format PGCIL-{ZONE}-YYYY-XXXXX
• Zone allocation corresponds to employee's current posting location
• Registered mobile number matches HRMS records
• Employee has active status (not on long leave or deputation)

Step 2: OTP Generation and Delivery
1. IT Support generates 8-digit alphanumeric OTP using PowerGrid SecureAuth application
2. OTP is sent to registered mobile number via SMS (OTP validity: 180 seconds or 3 minutes)
3. Employee must provide OTP within validity period to proceed

Step 3: Password Reset in SAP HR Module
Using SAP transaction ZPGHR001:
1. IT Support accesses the SAP HR password management module
2. Enters employee badge number and validates against HRMS database
3. Generates new temporary password conforming to PowerGrid security policy

Password Policy Requirements:
• Minimum length: 12 characters
• Must include: Uppercase letters, lowercase letters, numbers, and special characters (@#$%)
• Cannot contain: Employee ID, name, or common dictionary words
• Cannot reuse last 5 passwords
• Must be changed on first login within 24 hours

Step 4: SCADA Oracle Database Synchronization
Execute internal synchronization script to update SCADA credentials:
/usr/local/bin/sync_password.sh --employee-id {BADGE_NUMBER} --new-password {ENCRYPTED}

This script performs:
• Updates user credentials in SCADA_PROD Oracle database
• Synchronizes with PowerMIMIC HMI authentication system
• Updates LDAP directory for unified authentication
• Logs password change event to audit trail

Step 5: Incident Logging and Approval
1. Create ServiceNow incident ticket under category: ACCESS-PWD-RESET
2. Attach employee badge verification screenshot and OTP confirmation
3. Obtain digital approval from Zonal IT Head (auto-approved for standard resets)

4. Send password reset confirmation email to employee's registered email address

Typical Completion Time:
• Normal business hours: 15-20 minutes after receiving all required information
• After business hours: 30-45 minutes (on-call support engineer)
• Emergency cases affecting grid operations: Immediate priority (within 10 minutes)

4. Escalation Matrix and Support Contacts

L1 Support - 24x7 IT Helpdesk:
• Toll-Free Phone: 18001234321
• WhatsApp Support: +91-98765-43210 (Duty Engineer)
• Email: scada-support@powergrid.in
• Response Time SLA: 15 minutes for acknowledgment, 2 hours for resolution

L2 Support - SCADA Subject Matter Experts:
• Contact when L1 cannot resolve within 2 hours
• Email: scada-sme@powergrid.in
• Escalation trigger: Critical grid operations impact or national emergency
• Available: Monday-Friday 9 AM - 5 PM (extended hours for critical issues)

Emergency Escalation Chain:
1. Chief Manager (IT-SCADA): +91-98765-00001 - For zonal-level SCADA failures
2. General Manager (IT): +91-98765-00002 - For multi-zonal coordination required
3. Director (IT): +91-98765-00003 - Only for Grid Security incidents or National Emergency

Specialized Support Teams:
• Cybersecurity Incidents: cert-pg@powergrid.in (CC: cert-in@cert-in.org.in)
• Database Issues: dba-team@powergrid.in
• Network Problems: network-ops@powergrid.in
• SAP Module Issues: sap-support@powergrid.in

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